ALPHA MICROSYSTEMS
RIGHT. FROM THE START.

AlphaCALC®
Installation and Administrator's Guide

DSM-00086-00

© 1995 Alpha Microsystems

REVISIONS INCORPORATED	
REVISION	DATE
A00	October 1991

AlphaCALC Installation and Administrator's Guide

To re-order this document, request part number DS0-00086-00

The information contained in this manual is believed to be accurate and reliable. However, no responsibility for the accuracy, completeness or use of this information is assumed by Alpha Microsystems.

This document may contain references to products covered under U.S. Patent Number 4,530,048.

The following are registered trademarks of Alpha Microsystems, Santa Ana, CA 92799:

AMIGOS	AMOS	Alpha Micro	AlphaACCOUNTING
AlphaBASIC	AlphaCALC	AlphaCOBOL	AlphaDDE
AlphaFORTRAN 77	AlphaLAN	AlphaLEDGER	AlphaMAIL
AlphaMATE	AlphaNET	AlphaPASCAL	AlphaRJE
AlphaWRITE	CASELODE	OmniBASIC	VER-A-TEL
VIDEOTRAX			

The following are trademarks of Alpha Microsystems, Santa Ana, CA 92799:

AlphaBASIC PLUS	AlphaVUE	AM-PC	AMTEC
DART	ESP	MULTI	*in*Sight/am
*in*Front/am			

All other copyrights and trademarks are the property of their respective holders.

ALPHA MICROSYSTEMS
2722 S. Fairview St.
P.O. Box 25059
Santa Ana, CA 92799

TABLE OF CONTENTS

CHAPTER ONE - INTRODUCTION.....1-1
Who This Manual is For.....1-1
Graphic Conventions.....1-2

CHAPTER TWO - INSTALLATION.....2-1
Single Version Installation.....2-1
Step One: Define Ersatz Names and Reboot.....2-2
Step Two: Copy the Software.....2-2
Step Three: Verify the Software Files.....2-2
Verifying ESP and DART.....2-3
Step Four: Move Files to the Correct Accounts.....2-3
Step Five: Modify Your System .INI File.....2-4
Step Six: Reboot Using the Test .INI File.....2-5
Step Seven: Enter the Product Installation Code.....2-5
Step Eight: Defining Printers to AlphaCALC.....2-5
Dual Version Installation.....2-6
Step One: Define Ersatz Names and Reboot.....2-6
Step Two: Copy the Software.....2-7
Step Three: Verify the Software Files.....2-7
Verifying ESP and DART.....2-8
Step Four: Move Files to the Correct Accounts.....2-8
Step Five: Modify Your System .INI File.....2-9
Step Six: Reboot Using the Test .INI File.....2-10
Step Seven: Enter the Product Installation Code.....2-10
Step Eight: Defining Printers to AlphaCALC.....2-10
For ESP Users: Modifying the TOOLBX.INI File.....2-11

CHAPTER THREE - THE AlphaCALC INITIALIZATION FILE.....3-1
Initialization File Names and Locations.....3-1
The Default Initialization File.....3-2
Initialization File Commands.....3-2
Character Setting Commands.....3-2
The CUBE-WARNING Command.....3-3
The BORDERCOLOR Command.....3-3
The MAX-GRAPH-POINTS Command.....3-4
The GRAPHICS-DEVICE Command.....3-4

CHAPTER THREE - THE AlphaCALC INITIALIZATION FILE (continued)
The PRINTER Command..... 3-4
PRINTER Command Format..... 3-4
The FORMAT-MASK Command..... 3-5
The Format Definition.....3-6
FORMAT-MASK Samples..... 3-7

CHAPTER FOUR - MODIFYING AlphaCALC's SCREENS AND HELP FILES..... 4-1
Language File Selection..... 4-1
Modifying AlphaCALC'S Screens..... 4-2
The Modification Process.....4-2
Changing the Screen Title.....4-4
Screen File Names..... 4-5
Changing AlphaCALC'S Help Files..... 4-6
Procedure for Changing the Help Files.....4-7
Help File Names..... 4-7
Main Menu..... 4-8
Editing Menu..... 4-8
Worksheet Display..... 4-9
Modifying the Message File.....4-9
Changing the Messages..... 4-9
Translating the Message File.....4-10
Assembling the Message File.....4-10

CHAPTER FIVE - GRAPHICS SETUP.....5-1
Install AMIGOS.....5-1
Modifying the System Initialization File.....5-2
Defining the Available Graphics Devices.....5-2
The MAX-GRAPH-POINTS Command and Memory Usage..... 5-3

CHAPTER SIX - AlphaCALC's MEMORY USAGE..... 6-1
What AlphaCALC Uses Memory For.....6-1
Reducing AlphaCALC's Memory Usage.....6-1
ESP Toolbox Memory Use.....6-2
Toolbox per User Memory.....6-2
Sort and the Toolbox Memory Pool.....6-3

CHAPTER SEVEN - FILES ALPHACALC CREATES.....7-1
Backup Files..... 7-1
Print Files.....7-1
Directory Print Files.....7-2
Export Files.....7-2
Graph Files..... 7-2
AlphaWRITE Document Files..... 7-3
Macro Files..... 7-3

CHAPTER EIGHT - EXTENDED CHARACTER SUPPORT..... 8-1

APPENDIX A - FILE LIST A-1

DOCUMENT HISTORY

INDEX

READER'S COMMENTS

CHAPTER ONE
INTRODUCTION

This manual is for the AlphaCALC administrator—the person whose job it is to install AlphaCALC, configure it properly for your computer, and keep it up and running.

This book's chapters cover these topics:

- Chapter One gives a brief overview of the book and describes the graphic conventions it uses.
- Chapter Two tells you how to install AlphaCALC.
- Chapter Three describes the AlphaCALC initialization file. The initialization file lets you tailor AlphaCALC to how you want it to run— you can specify the printers available, create new cell formats, and set various other options.
- Chapter Four tells you how to modify AlphaCALC's screen displays.
- Chapter Five covers setting up AlphaCALC's graphics capabilities.
- Chapter Six discusses AlphaCALC memory usage and how you can reduce AlphaCALC's user memory requirements.
- Chapter Seven describes the different types of files AlphaCALC's various features can create.
- Chapter Eight covers how you can use extended (non-ASCII) characters in AlphaCALC.

In addition, an appendix describes the files included with the AlphaCALC release.

WHO THIS MANUAL IS FOR

This manual is for the person in charge of AlphaCALC on your computer. If possible, this person should be an experienced AMOS user. We assume you (our reader) have a certain level of knowledge about your computer. In particular, you should be familiar with:

- Installing software from a VCR tape.
- Using the AlphaVUE text editor.
- Modifying the system initialization command file.

Any time we discuss an aspect of your computer you aren't familiar with, you can refer to one or more of these other Alpha Micro publications for more information:

- The *Owner's Manual* for your computer.
- The *AMOS System Commands Reference Manual*.
- The *AMOS System Operator's Guide*.
- The *AMOS System Operator's Guide to the System Initialization Command File*.

GRAPHIC CONVENTIONS

In order to make this document as easy as possible to understand, we use several conventions in layout and type style, so you can recognize whether we are talking about a command or key, when you're to type an entry, and so on.

CONVENTION	EXPLANATION
COMMAND	AMOS commands and file names, such as CALC.INI, are shown in all capitals.
AlphaCALC Main Menu	Screen names are printed in the same type as regular text, with initial capitals.
`What You Type`	Text you are expected to type is shown in a bold Courier typeface.
`What AlphaCALC Displays`	Text displayed on your screen is shown in Courier typeface.
PRINT	AlphaCALC command names are shown in small, bold capitals.
KEY	Keys you need to press are shown in a keycap box.
	The halt symbol marks important information you should read carefully before continuing. This often indicates something you MUST or MUST NOT do, so read carefully.
	The hint symbol marks a helpful bit of information. It often points out a short cut which may save you time or trouble.
	The remember symbol marks information you need to keep in mind while you are following a set of instructions.

CHAPTER TWO
INSTALLATION

This chapter tells you how to install AlphaCALC on your computer. There are two different installation procedures in this chapter:

- The first one tells you how to install AlphaCALC 2.0 so it is the only version of AlphaCALC on your computer. Use this procedure if you don't have an earlier version of AlphaCALC, or if you want to use only AlphaCALC 2.0.

- The second procedure lets you install AlphaCALC 2.0 "side-by-side" with an earlier version of AlphaCALC. Use this procedure if you want to be able to use both AlphaCALC 2.0 and an earlier version of AlphaCALC on the same computer. If you want to do this, see the *AlphaCALC 2.0 Release Notes* for important compatibility information.

SINGLE VERSION INSTALLATION

To install AlphaCALC 2.0 as the only version of AlphaCALC on your computer requires the following procedure. Each step is covered in more detail below.

1. Decide what accounts you want to keep the AlphaCALC software in. Define ersatz names for these accounts.

2. Copy the software from the release tape to your hard disk.

3. Verify that all files were copied correctly.

4. Move some files to the correct accounts.

5. Modify your system initialization command (.INI) file.

6. Reboot using the test .INI file.

7. Call up AlphaCALC and enter the Product Installation Code (PIC).

8. Define the printers you want to use in AlphaCALC.

After completing this procedure, you'll be able to use all of AlphaCALC except the graph generating features. To activate graphics, follow the instructions in Chapter 5.

STEP ONE: DEFINE ERSATZ NAMES AND REBOOT

The bulk of the AlphaCALC software is contained in two different accounts. You need to decide what device you want these accounts on, and assign ersatz names to the accounts.

The main AlphaCALC software account must have the ersatz name ALC:. The default for this account is DSK0:[7,13], but you can assign this name to any account you want.

Some AlphaCALC files (these files are part of ESP, the software which controls the AlphaCALC screen displays) go in an account with the ersatz name DMSESP:. This account can also be on any device. Its default is DSK0:[7,32].

If you use the ESP package, the DMSESP: account is already defined. Use this account during the AlphaCALC installation.

If you do not use ESP, you may want to define the ALC: and DMSESP: ersatz names as the same disk account. Then, in step four, below, you do not need to perform the file copies from ALC: to DMSESP:, since they are the same account.

Decide what accounts you want to keep the AlphaCALC software in, then add statements defining ALC: and DMSESP: to your ersatz definition file.

After updating the ersatz file, reboot your computer so the definitions take affect.

STEP TWO: COPY THE SOFTWARE

In this step, you copy all the files from the AlphaCALC release tape to the ALC: account. Later, you move some files to their proper accounts.

Use these commands to download the software:

```
LOG ALC: RETURN
VCRRES =ALL:*.*[] RETURN
```

This copies all the files to the ALC: account.

STEP THREE: VERIFY THE SOFTWARE FILES

The file CALC.DIR is copied to the ALC: account as part of the AlphaCALC software. This file contains names, versions (where applicable) and hash totals for all the files comprising AlphaCALC, formatted to use with the VERIFY program.

If you've chosen an account other than DSK0:[7,13] as ALC:, you need to change the device and account specifications in the CALC.DIR file before you run VERIFY. Use AlphaVUE to modify the file. VERIFY doesn't understand ersatz names.

When you modify CALC.DIR, be sure you don't change the column the version numbers and hash totals start in. These numbers must be in the correct columns of the file for VERIFY to work.

Now, run VERIFY:

VERIFY CALC RETURN

VERIFY lists each file on your screen as it checks it. The message `VERIFIED` should appear next to each file.

If VERIFY shows any problems, repeat step two, making sure you're in the correct account when you use the VCRRES command. If this doesn't correct the problem, contact your Alpha Micro dealer.

If you do not have the ESP or DART software package installed on your computer, continue with step four.

Verifying ESP and DART

If you have ESP and/or DART installed on your computer, you need to check their versions before continuing. Only ESP and DART versions 2.0 or later, supplied by Alpha Micro, are compatible with AlphaCALC. AlphaCALC includes some of the same files as ESP, and copying AlphaCALC over an earlier version of ESP may make ESP or DART unusable.

To let you make sure your software is compatible, AlphaCALC includes DIR files for ESP and DART (ESP.DIR and DART.DIR). If necessary, modify these files as described above for CALC.DIR, entering the correct account numbers, then run VERIFY for ESP (and DART if you have it):

VERIFY ESP RETURN
VERIFY DART RETURN

If any of the listed files do not verify, there is a possible compatibility problem. Contact your Alpha Micro dealer.

STEP FOUR: MOVE FILES TO THE CORRECT ACCOUNTS

Now, you'll take some of the AlphaCALC files and copy them to accounts other than ALC:. To do this, first log into OPR:, then use this series of commands:

As mentioned in step one, if you define ALC: and DMSESP: as the same disk account, you do not have to enter the commands below which begin **COPY DMSESP:=**.

```
COPY SYS:=ALC:CALC.LIT RETURN
COPY DVR:=ALC:WRITE.PDV RETURN
COPY BAS:=ALC:EXTCEL.SBR,EXTCEL.BSI RETURN
COPY MAC:=ALC:EXTCEL.LIB,EXTCEL.CPY RETURN
COPY DMSESP:=ALC:TOOLBX.*,DAC*.*,DAS*.* RETURN
COPY DMSESP:=ALC:OPNBOX.USA,PRTSEL.USA,GRFSEL.USA RETURN
COPY DMSESP:=ALC:TBXMSG.USA,PSSWRD.USA RETURN
```

If your computer is running a 1.X version of AMOS, enter these commands:

```
COPY SYS:AMSORT.SYS=ALC:AMSORT.14 RETURN
COPY BAS:BASORT.SBR=ALC:BASORT.14 RETURN
```

If your computer uses an AMOS 2.X operating system, type these commands:

```
COPY SYS:AMSORT.SYS=ALC:AMSORT.22 RETURN
COPY BAS:BASORT.SBR=ALC:BASORT.22 RETURN
COPY BP:BASORT.XBR=ALC:BASORT.22 RETURN
COPY BP:EXTCEL.XBR=ALC:EXTCEL.SBR RETURN
```

The BP: account is for AlphaBASIC PLUS files. If you do not use AlphaBASIC PLUS, you may not have a BP: account. In that case, omit those commands.

If you do not use ESP, enter these commands:

```
LOG DMSESP: RETURN
RENAME TOOLBX.INI=TOOLBX.NEW RETURN
```

If you do use ESP, see the section on the TOOLBX.INI file at the end of this chapter.

STEP FIVE: MODIFY YOUR SYSTEM .INI FILE

Next, you need to make some additions to your system initialization command file. This file is in SYS: and is called AMOS32.INI on AMOS/32 systems and AMOSL.INI on AMOS/L systems. We'll refer to it as the .INI file.

NEVER modify the system .INI file directly. Always make a copy and modify the copy. If you make an error in the original .INI file it could keep your computer from booting!

AlphaCALC needs some files loaded in system memory in order to function. Add these statements to the SYSTEM section of your .INI file:

```
SYSTEM AMSORT.SYS
SYSTEM DMSESP:TOOLBX.SYS/N
SYSTEM AMIGOS.SYS/N
SYSTEM GRAPH.SYS
```

If the DMSESP: account is not on DSK0:, you must MOUNT the device it's on in the .INI file, before the SYSTEM statements.

The last two statements above, the ones loading AMIGOS.SYS and GRAPH.SYS, are required for AMIGOS, Alpha Micro's graphics operating system, which AlphaCALC uses to generate and display graphs. If you've installed AMIGOS, these statements should already be in the .INI file. If you don't have the AMIGOS software, you may want to mark these statements as comments, or leave them out altogether. If you leave the statements in, they will cause error messages during bootup, but no other problems. See Chapter 5 for more information on AMIGOS and setting up AlphaCALC's graphics feature.

STEP SIX: REBOOT USING THE TEST .INI FILE

After adding the necessary statements to the .INI file, finish from the file. Make sure no one is using the computer, and you've disconnected from all networks. Then, MONTST using your test .INI file.

If the MONTST works and the correct files are now loaded in system memory, you can rename the test .INI file to the correct name for your initialization file.

STEP SEVEN: ENTER THE PRODUCT INSTALLATION CODE

Like all Alpha Micro software, AlphaCALC is protected by a Software Security Device (SSD). Your SSD is unique to your computer, and requires you to enter a Product Installation Code (PIC). If you don't have your PIC, you can get it from your dealer.

To enter the AlphaCALC PIC, type this command:

CALC [RETURN]

You'll see a screen requesting the PIC. Type it in, being careful to use the correct capitalization and punctuation. After a few second delay—during which time all jobs on your system will be suspended—you'll see the AlphaCALC Main Menu.

If, after installation, you need to see your PIC, type this command:

CALC/P [RETURN]

STEP EIGHT: DEFINING PRINTERS TO AlphaCALC

In order to print from AlphaCALC, you need to tell AlphaCALC what type of printers are available to it. You can do this either in the system printer definition file, PRTDEF, or in the AlphaCALC initialization file. See Chapter 3 for the correct way to define printers you want to use from AlphaCALC.

DUAL VERSION INSTALLATION

To install AlphaCALC 2.0 on your computer alongside an earlier version of AlphaCALC, so you can use either version, requires the following procedure. Each step is covered in more detail below.

While this general outline is the same as for the single version installation, the detailed information in the steps is different. The two installation processes are NOT interchangeable!

1. Decide what accounts you want to keep the AlphaCALC software in. Define ersatz names for these accounts.
2. Copy the software from the release tape to your hard disk.
3. Verify that all files were copied correctly.
4. Move some files to the correct accounts.
5. Modify your system initialization command (.INI) file.
6. Reboot using the test .INI file.
7. Call up AlphaCALC and enter the Product Installation Code (PIC).
8. Define the printers you want to use in AlphaCALC.

After completing this procedure, you'll be able to use all of AlphaCALC except the graph generating features. To activate graphics, follow the instructions in Chapter 5.

STEP ONE: DEFINE ERSATZ NAMES AND REBOOT

The bulk of the AlphaCALC software is contained in two different accounts. You need to decide what device you want these accounts on, and assign ersatz names to the accounts.

The main AlphaCALC software account must have the ersatz name ALC:. Since older versions of the AlphaCALC software keep their files in DSK0:[7,13], you **cannot** use this account for AlphaCALC 2.0. You can use any other device and account you want.

Some AlphaCALC files (these files are part of ESP, the software which controls the AlphaCALC screen displays) go in an account with the ersatz name DMSESP:. This account can also be on any device. Its default is DSK0:[7,32].

If you do not use ESP, you may want to define the ALC: and DMSESP: ersatz names as the same disk account. Then, in step four, below, you do not need to perform the file copies from ALC: to DMSESP:, since they are the same account.

If you use the ESP package, the DMSESP: account is already defined. Use this account during the AlphaCALC installation.

This procedure lets you use two versions of AlphaCALC on your computer at once. It does not let you use two versions of ESP. If you have an earlier version of ESP on your computer, it is overwritten during installation! See step three, below.

Decide what accounts you want to keep the AlphaCALC software in, then add statements defining ALC: and DMSESP: to your ersatz definition file.

After updating the ersatz file, reboot your computer so the definitions take affect.

STEP TWO: COPY THE SOFTWARE

All the software on the AlphaCALC release tape is in a single account. You copy all the files to the ALC: account, then, later, move some files to their proper accounts.

Use these commands to download the software:

```
LOG ALC: RETURN
VCRRES =ALL:*.*[] RETURN
```

This copies all the files to the ALC: account.

STEP THREE: VERIFY THE SOFTWARE FILES

The file CALC.DIR is copied to the ALC: account as part of the AlphaCALC software. This file contains names, versions (where applicable) and hash totals for all the files comprising AlphaCALC, formatted to use with the VERIFY program.

Since the ALC: account is not DSK0:[7,13], you need to change the device and account specifications in the CALC.DIR file before you run VERIFY. Use AlphaVUE to modify the file. VERIFY doesn't understand ersatz names.

When you modify CALC.DIR, be sure you don't change the column the version numbers and hash totals start in. These numbers must be in the correct columns of the file for VERIFY to work.

Now, run VERIFY:

```
VERIFY CALC RETURN
```

VERIFY lists each file on your screen as it checks it. The message `VERIFIED` should appear next to each file.

If VERIFY shows any problems, repeat step two, making sure you're in the correct account when you use the VCRRES command. If this doesn't correct the problem, contact your Alpha Micro dealer.

If you do not have the ESP or DART software packages installed on your computer, continue with step four.

Verifying ESP and DART

If you have ESP and/or DART installed on your computer, you need to check their versions before continuing. Only ESP and DART versions 2.0 or later, supplied by Alpha Micro, are compatible with AlphaCALC. AlphaCALC includes some of the same files as ESP, and copying AlphaCALC over an earlier version of ESP may make ESP or DART unusable.

To let you make sure your software is compatible, AlphaCALC includes DIR files for ESP and DART (ESP.DIR and DART.DIR). If necessary, modify these files as described above for CALC.DIR, entering the correct account numbers, then run VERIFY for ESP (and DART if you have it):

```
VERIFY ESP RETURN
VERIFY DART RETURN
```

If any of the listed files do not verify, there is a possible compatibility problem. Contact your Alpha Micro dealer.

STEP FOUR: MOVE FILES TO THE CORRECT ACCOUNTS

Now, you'll take some of the AlphaCALC files and copy them to accounts other than ALC:. To do this, first log into OPR:, then use this series of commands:

As mentioned in step one, if you define ALC: and DMSESP: as the same disk account, you do not need to enter the commands below which begin **COPY DMSESP:**.

```
COPY DVR:=ALC:WRITE.PDV RETURN
COPY BAS:=ALC:EXTCEL.SBR,EXTCEL.BSI RETURN
COPY MAC:=ALC:EXTCEL.LIB,EXTCEL.CPY RETURN
COPY DMSESP:=ALC:TOOLBX.*,DAC*.*,DAS*.* RETURN
COPY DMSESP:=ALC:OPNBOX.USA,PRTSEL.USA,GRFSEL.USA RETURN
COPY DMSESP:=ALC:TBXMSG.USA,PSSWRD.USA RETURN
```

If your computer is running a 1.X version of AMOS, enter these commands:

```
COPY SYS:AMSORT.SYS=ALC:AMSORT.14 RETURN
COPY BAS:BASORT.SBR=ALC:BASORT.14 RETURN
```

If your computer uses an AMOS 2.X operating system, type these commands:

```
COPY SYS:AMSORT.SYS=ALC:AMSORT.22 RETURN
COPY BAS:BASORT.SBR=ALC:BASORT.22 RETURN
COPY BP:BASORT.XBR=ALC:BASORT.22 RETURN
COPY BP:EXTCEL.XBR=ALC:EXTCEL.SBR RETURN
```

The BP: account is for AlphaBASIC PLUS files. If you do not use AlphaBASIC PLUS, you may not have a BP: account. In that case, omit those commands.

If you do not use ESP, enter these commands:

```
LOG DMSESP: RETURN
RENAME TOOLBX.INI=TOOLBX.NEW RETURN
```

If you do use ESP, see the section on the TOOLBX.INI file at the end of this chapter.

You also need to copy the main AlphaCALC program file, CALC.LIT, to the SYS: account, so you can call up AlphaCALC 2.0 from any account. But, you **must** give this file a different name, so it doesn't overwrite the CALC.LIT file of your earlier version of AlphaCALC.

We suggest you use the name ACALC for this file. The command to use would then be:

```
COPY SYS:ACALC.LIT=ALC:CALC.LIT RETURN
```

The name you give the CALC.LIT file will be the command you use to call up AlphaCALC 2.0.

STEP FIVE: MODIFY YOUR SYSTEM .INI FILE

Next, you need to make some changes to your system initialization command file. This file is in SYS: and is called AMOS32.INI on AMOS/32 systems and AMOSL.INI on AMOS/L systems. We'll refer to it as the .INI file.

NEVER modify the system .INI file directly. Always make a copy and modify the copy. If you make an error in the original .INI file it could keep your computer from booting!

AlphaCALC needs some files loaded in system memory in order to function. Add these statements to the SYSTEM section of your .INI file:

```
SYSTEM AMSORT.SYS
SYSTEM DMSESP:TOOLBX.SYS/N
SYSTEM AMIGOS.SYS/N
SYSTEM GRAPH.SYS
```

If the DMSESP: account is not on DSK0:, you must MOUNT the device it's on in the .INI file, before the SYSTEM statements.

The statements loading AMIGOS.SYS and GRAPH.SYS are required for AMIGOS, Alpha Micro's graphics operating system, which AlphaCALC uses to generate and display graphs. If you've installed AMIGOS, these statements should already be in the .INI file. If you don't have the AMIGOS software, you may want to mark these statements as comments, or leave them out altogether. If you leave the statements in, they will cause error messages during bootup, but no other problems. See Chapter 5 for more information on AMIGOS and setting up AlphaCALC's graphics feature.

In addition, to avoid conflicts between your two versions of AlphaCALC, you must make sure that certain files are **not** loaded in system memory. If there are SYSTEM statements for any of the following files, remove them:

ALC*.OVR
ALC*.USA
CH*.USA

STEP SIX: REBOOT USING THE TEST .INI FILE

After adding and removing the necessary statements in the .INI file, finish from the file. Make sure no one is using the computer, and you've disconnected from all networks. Then, MONTST using your test .INI file.

If the MONTST works and the correct files are now loaded in system memory, you can rename the test .INI file to the correct name for your initialization file.

STEP SEVEN: ENTER THE PRODUCT INSTALLATION CODE

Like all Alpha Micro software, AlphaCALC is protected by a Software Security Device (SSD). Your SSD is unique to your computer, and requires you to enter a Product Installation Code (PIC). If you don't have your PIC, you can get it from your dealer.

To enter the AlphaCALC PIC, type this command:

ACALC [RETURN]

If you gave the CALC.LIT file a name other than ACALC in step 4, use that name here.

You'll see a screen requesting the PIC. Type it in, being careful to use the correct capitalization and punctuation. After a few second delay—during which time all jobs on your system will be suspended—you'll see the AlphaCALC Main Menu.

After installation, if you need to know your PIC, you can display it by entering this command:

ACALC/P [RETURN]

If you call up AlphaCALC 2.0 by entering something other than ACALC, use that command here.

STEP EIGHT: DEFINING PRINTERS TO AlphaCALC

In order to print from AlphaCALC, you need to tell AlphaCALC what type of printers are available to it. You can do this either in the system printer definition file, PRTDEF, or in the AlphaCALC initialization file. See Chapter 3 for the correct way to define printers you want to use from AlphaCALC.

FOR ESP USERS: MODIFYING THE TOOLBX.INI FILE

Because AlphaCALC uses the ESP toolbox, it requires a TOOLBX.INI file. A file configured for AlphaCALC is included on the release tape, with the name TOOLBX.NEW. If you use ESP, you already have a TOOLBX.INI file, and you must check this file to make sure it is compatible with AlphaCALC.

This is what to look for in your TOOLBX.INI file:

- The Macros desk accessory feature, DACMAC, must be set up. The statement to do so is:

  ```
  ACCESSORY=DMSESP:DACMAC
  ```

- There must be a MEMORY statement setting the toolbox memory pool size to at least 16K.

As long as your TOOLBX.INI file contains these two statements, all of AlphaCALC's functions should work.

Only the Macros desk accessory is installed with AlphaCALC. If you have other ESP accessories installed, your desk accessory menu will appear when you press AlphaCALC's [MACROS] key.

CHAPTER THREE
THE AlphaCALC INITIALIZATION FILE

This chapter describes the AlphaCALC initialization file. The initialization file is a text file you can create and edit using AlphaVUE. It lets you set various options which affect the way AlphaCALC starts and behaves once it's running. You can define the available printers and graphics devices, define additional cell formats, and more.

INITIALIZATION FILE NAMES AND LOCATIONS

You can have more than one AlphaCALC initialization file on your computer. If you do, which file AlphaCALC looks at when you call it up depends on the language you use and the disk account you're in. This lets you tailor AlphaCALC to individual users and to the functions performed in various projects and accounts on your disk.

The initialization file can be called either CALC.INI or CALC.lan, where "lan" is the language extension defined in the language definition file currently selected for your job (for example, CALC.USA). This lets people using different languages on the same computer have different versions of the initialization file, even if they're working in the same accounts. If everyone on your computer uses the same language, you would normally call the initialization file CALC.INI.

In order to be as flexible as possible, AlphaCALC goes through a fairly involved search procedure when scanning for the initialization file. It searches disk accounts in this order:

1. Your current account.
2. The project library ([,0]) account.
3. The ALC: account.

If it finds a CALC.lan file for your selected language in any of these accounts, it stops looking. If it doesn't find CALC.lan in any of these places, it repeats the search path, looking for CALC.INI.

Note that a language specific initialization file in any account in the search path has precedence over any CALC.INI file. For example: you're in account [152,15], and your language extension is UK. If there's a CALC.UK file in ALC:, it will be used as your initialization file, even if there's a CALC.INI file in [152,15].

Through the rest of this chapter, we refer to the AlphaCALC initialization file as CALC.INI. Anything we say about CALC.INI is also true for a CALC.lan initialization file, unless stated otherwise.

THE DEFAULT INITIALIZATION FILE

A default AlphaCALC initialization file in included with the AlphaCALC software. This file is called CALC.NEW, and it's in the ALC: ersatz account. You can copy this file to the account where you want an initialization file—giving it the correct extension—and use it as a basis for your file.

INITIALIZATION FILE COMMANDS

The commands available in the initialization file are described below. All the commands, except the PRINTER command, are in the following format:

COMMAND=ARGUMENT(s)

The PRINTER command also consists of a command followed by arguments, but it does not include the equal sign.

All the sample commands in this chapter are shown without spaces on the command line. If you want, you can include spaces before and/or after the equal sign.

In addition to these commands, you can include comments in CALC.INI. Any line which begins with a semicolon (;) is a comment, and AlphaCALC ignores it.

CHARACTER SETTING COMMANDS

CALC.INI can contain several commands setting the character(s) you enter and/or AlphaCALC displays at various times during operation. For example, you can change the character used to mark an absolute cell reference. The table below gives the command name, the default value, and a short description for each of these commands.

To change any of these characters, enter a line in this format in CALC.INI:

```
COMMAND=character
```

Except where noted, each command accepts only a single character as its argument.

COMMAND	DEFAULT	DESCRIPTION
OVERFLOWCHARACTER	#	The character that displays before a number when the number is too long to fit in its cell.
ABSOLUTE-INDICATOR	$	The character used to indicate a cell reference in a cell formula is absolute. Absolute references are not changed when the cell is copied.
RANGE-SEPARATOR	:	The character you enter (and AlphaCALC displays) to separate the beginning and ending cells in a cell range. This command affects cell ranges everywhere you use them in AlphaCALC, both in cell formulas and on command screens. It accepts multiple characters.
LAYER-SEPARATOR	|	The character you use to separate the layer number from the row letter when you enter a cell ID.

THE CUBE-WARNING COMMAND

The CUBE-WARNING command can help prevent you from inadvertently modifying other layers of your worksheet. When CUBE-WARNING is set to true, AlphaCALC displays a warning message whenever you request a feature that will affect multiple layers of your worksheet—for example, copying or erasing a marked block that includes cells on multiple layers.

The default for CUBE-WARNING is true; if you don’t want to see the warning message, set CUBE-WARNING=FALSE.

THE BORDERCOLOR COMMAND

BORDERCOLOR lets you set the colors of the row and column borders of your worksheet. If you specify only one color number, it is used as the foreground color. You can set both foreground and background color by separating the numbers with a slash (/). For example:

```
BORDERCOLOR=7/3
```

This sets the foreground color to 7 and the background to 3.

The default setting is 6/0.

THE MAX-GRAPH-POINTS COMMAND

The MAX-GRAPH-POINTS command sets the maximum number of points you can have in any single cell range of a graph. The default value is 40.

This command determines the amount of memory AlphaCALC reserves for graph data when you press the [GRAPH SETUP] key. For more information on memory use during graphics, see Chapter 5.

THE GRAPHICS-DEVICE COMMAND

The GRAPHICS-DEVICE command lets you define the devices available for displaying or printing an AlphaCALC generated graph. It is described in Chapter 5.

THE PRINTER COMMAND

You can define up to sixteen printers to use with AlphaCALC. The definitions can be in either of two places:

- The AlphaCALC initialization file.
- The system printer definition file, PRTDEF, in SYS: (DSK0:[1,4]). Like the AlphaCALC initialization file, PRTDEF can have the extension .INI or an extension based on your language.

Each location has advantages: using the system printer definition file lets you define all the printers for all users and multiple software packages in one place, while using CALC.INI lets you customize the printer selections for each account where AlphaCALC is used.

AlphaCALC looks in the current AlphaCALC initialization file for printer definitions first—if there are any PRINTER statements, AlphaCALC uses these definitions. If there are no printers defined in the current CALC.INI, AlphaCALC looks in PRTDEF.

This means you cannot combine printer definition files, listing some printers in PRTDEF and some in CALC.INI. If there are any printers defined CALC.INI, AlphaCALC will not take any definitions from PRTDEF.

PRINTER Command Format

Whether in PRTDEF or in CALC.INI, each available printer is defined by a single line PRINTER statement, in this format:

```
PRINTER name,driver,"description",,,cpuid
```

Here is a sample printer definition:

```
PRINTER LASWR1,LASWRT,"Personnel Laser Printer",,,14982375-
```

The parameters in the PRINTER statement must be separated by commas, with no spaces except the one after PRINTER. The parameters are:

NAME — The name of the printer, as defined to your printer spooler. In the sample, this is LASWR1.

DRIVER — The name of the printer driver to use with this printer. This is LASWRT in the sample. AlphaCALC is designed to support any printer driver which conforms to Alpha Micro standards.

DESCRIPTION — This is the description of the printer you'll see when you use AlphaCALC. It can be up to 30 characters long, and must be enclosed in double quotes, as shown. The sample description is Personnel Laser Printer.

CPUID — If the printer is on another computer which your computer is attached to using AlphaNET, this is the CPUID of the computer the printer is physically attached to. You can use either the numeric CPUID, as shown, or, if one is set up, the ersatz name of the other computer.

If the printer is attached to this computer, you can omit the CPUID.

There must be three commas between the description and the CPUID. These commas are "placeholders" for printer parameters which are used by other applications but which AlphaCALC does not support.

THE FORMAT-MASK COMMAND

You can use the FORMAT-MASK command to define custom cell formats in addition to AlphaCALC's standard formats. The standard available formats are described in the *AlphaCALC Reference Book*. If you want to use a different format—for example, if you want to display values as currency with three decimal places—use FORMAT-MASK.

You can define only numeric formats with FORMAT-MASK. You cannot create new date formats, and, like the standard formats, custom formats do not affect text strings.

Here is the overall format for this command:

```
FORMAT-MASK=code,type,format,negative format,zero format
```

As this shows, you can actually define up to three separate cell formats for each format code: one for positive values, one for negative values, and one for zero values.

The parts of FORMAT-MASK are:

CODE — The one to three character code you'll enter in AlphaCALC to choose this format. This must be unique for each format.

TYPE — The data type this format is to be used with. This should always be 0, for numeric. Other settings are reserved for future development.

FORMATS The positive, negative and zero format sections are each subdivided into several areas. They are discussed in detail below.

The Format Definition

The positive, negative, and zero formats each have this structure:

```
"mask",0,0,digits,scaling,0,0
```

Each FORMAT-MASK statement must have a positive format. If you do not enter a negative format, negative values use the positive format with a minus sign in front of the value. If there is no zero format, zero is displayed using the positive format.

The parts of the structure are:

MASK This is the actual format you want the data to display in. The mask format used is similar to AlphaBASIC's. For numbers, you can use these characters:

# = a digit, do not display leading zeros.
0 = a digit, display leading zeros.

For example, the number 12 will display as 12 if its mask is "###", and as 012 if the mask is "000". You can combine these characters if you want only a certain number of leading zeros. For example, "#00.00" is a valid mask.

If you want a specific number of decimal places, include the decimal point, with the proper number of 0's or #'s after it. You must also set "digits" to the number of decimal places, as described below.

You can also include text characters, such as letters, dollar signs, commas, or parentheses, at the appropriate places in the mask. Certain characters have special meaning in a mask:

$ The currency symbol from the currently selected language definition file.
. The decimal point from the currently selected language definition file.
, The thousands separator from the currently selected language definition file.
E Indicates an exponent, as in scientific notation. This is true for capital E only; you can use lower case 'e' in text in the mask.

0,0 The next two places are reserved for future use.

DIGITS The number of decimal places—digits to the right of the decimal point—in the format.

SCALING	The scaling factor lets you multiply a value by a power of ten before display. For example, percentage formats multiply the value by 100 (a value of 1 equals 100%), and therefore have a scaling factor of 2 (100 equals 10 to the second power).
0,0	The last two places in the mask are reserved for future use. They should always contain zeros.

FORMAT-MASK Samples

The best way to explain what you can do using FORMAT-MASK is with an example. Here is a sample numeric format:

```
FORMAT-MASK=DC-,0,"$0.00",0,0,2,0,0,0,"-$0.00",0,0,2,0,0,0
```

This defines the DC- format—or "dollars and cents with minus sign." The format has these characteristics:

- The first 0 after DC- indicates it is for numeric values.
- There are separate masks defined for positive and negative values.
- The positive mask prints a dollar sign before the value, requires only one digit before the decimal point and rounds to and displays two decimal places.

Even though there’s only one digit shown left of the decimal, larger values will display correctly. The single 0 just indicates no other digits are required.

- The negative mask is the same as the positive, with the addition of a minus sign before the dollar sign.
- Both masks round to two decimal places and have no scaling factor.

Here is a more specialized example, one you could use in a price list:

```
FORMAT-MASK=UPR,0,"$0.00 each",0,0,2,0,0,0
```

This UPR—for "unit price"—format has many of the characteristics of our other sample. Amounts are preceded by a dollar sign, and have two decimal places, for dollars and cents. The differences are:

- The word "each" will display after the numeric value.
- There is no negative mask. Unit prices should always be positive. If there are any negative values, they will use the positive mask, with a minus sign preceding the value.

With the ability to use text in a format mask, you can create custom formats for your specialized needs.

If you transfer worksheets between accounts or between systems, be careful when using custom formats. If you use a custom format which exists only in a local CALC.INI file, the cells which use it will be changed to general format when you move the worksheet elsewhere.

CHAPTER FOUR
MODIFYING AlphaCALC'S SCREENS AND HELP FILES

This chapter describes how you can modify AlphaCALC screen displays, help files, and messages. Since these files are often modified for translation to languages other than American English, it begins with a discussion of how AlphaCALC determines what language to use.

LANGUAGE FILE SELECTION

The default language for an AMOS system is defined when you set up the system monitor. You can define multiple languages for one computer by loading the appropriate language definition files into system memory. Then, each user on the system can choose the language he or she wants to use by entering a SET LANG command.

When you call up AlphaCALC, it checks the language currently selected for your job, and uses the screen and help files with the correct extensions for that language. For example, if your language is American English, AlphaCALC uses screen and help files with the extension .USA.

The .USA files are the only ones supplied with AlphaCALC. You can create versions of these files for use with other languages by copying the files to the correct extensions, then following the procedures in this chapter for changing the text in them.

Creating versions of the AlphaCALC message file, ALCMSG, for other languages requires a different procedure. See the section "Modifying the Message File" at the end of this chapter.

You can have as many versions of these AlphaCALC files (with different extensions) as you want. AlphaCALC selects the ones to use based on the language setting for the current user. So, while you are working in AlphaCALC using English language screens, the person in the next office may be using French, Spanish, or so on.

Your language affects more than the screen files you use in AlphaCALC. AlphaCALC also reads your language definition file to determine your proper date format, decimal point and currency symbol, and yes/no responses. Also, see Chapter 3 for a discussion of language dependent AlphaCALC initialization files.

For more information on defining languages to AMOS, see the *AMOS System Operator's Guide*.

MODIFYING AlphaCALC'S SCREENS

The AlphaCALC release includes a program, ALCSCR, you can use to edit or translate any AlphaCALC screen display.

ALCSCR is a specially modified version of SEDIT, the screen editor portion of the Executive Screen Processor (ESP), the software package which was used to create all of AlphaCALC's screens. ALCSCR lets you change only those fields you can safely modify without interfering with AlphaCALC's operation.

Follow all the cautions below carefully! Modifying a screen in a way other than we describe here could cause very unpredictable results in AlphaCALC!

THE MODIFICATION PROCESS

Each AlphaCALC screen is contained in a separate file in the ALC: account. These files all have a name with the format ACSxxx.USA. A complete list of AlphaCALC screen files is given later in this chapter.

To change a screen display, you log into ALC: (or DMSESP: for the files found there), and call up ALCSCR, including the screen file name. For our example, we'll use the file ACSPST.USA, the American English file for the **PASTE** command screen:

ALCSCR ACSPST.USA [RETURN]

You now see the ALCSCR Main Menu. The edit option is the default, so press [RETURN] to view the Screen Editor display of the **PASTE** screen. This is what you see:

```
+----------  Paste in a worksheet into current AlphaCALC worksheet  -----------+
!                                                                               !
!                                                                               !
!                    Name of worksheet to paste with: ______                    !
!                                                                               !
!                                                                               !
!                                                                               !
!                                                                               !
!                                                                               !
!                                                                               !
!      [ Enter the name of the worksheet you want to copy into this one ]      !
!                                                                               !
!   [ UP ARROW or DOWN ARROW displays a directory of AlphaCALC worksheets ]    !
!                                                                               !
!          [ Press EXECUTE to perform the PASTE -OR- MENU to cancel ]           !
!                                                                               !
+-------------------------------------------------------------------------------+
```

To change the wording or color of any prompt or message on the screen, follow the steps below.

1. Move the cursor to the field you want to change, using the arrow keys. If you want to change the screen title, see the next section.

2. Press [F1] or [ESC] [E].

3. You now see the Field Editor screen, which shows all the information about this field. It looks like this (using the worksheet name field from the **PASTE** screen):

```
                              AlphaCALC Field Editor

                            Editing Field Number: 1
+-------------------------------------------------------------------------------+
!         Background color: 0                   Foreground color: 0             !
+-------------------------------------------------------------------------------+
!Field Buffer (text for screen titles):                                         !
!                                                                               !
!Field Default:                                                                 !
!_______________________________________________________________________________!
!Pop-Up Menu Selections:                                                        !
!                                                                               !
!Field Prompt:                                                                  !
!Name of worksheet to paste with:                                               !
!On-Line Help:                                                                  !
!Enter worksheet name -OR- press arrow keys to see directory.                   !
+-------------------------------------------------------------------------------+
         [ Press MENU key to cancel edit and return to screen display ]
               [ Press EXECUTE key to set new field information ]
          Enter the value to be used if NO data is entered by the user
```

The fields you can modify are:

- Foreground Color and Background Color: Leaving these set to 0, as shown above, uses the default foreground and background colors for your terminal. If you want to change this for some reason, enter a number from 1 to 7.

- Field Buffer (text for screen titles): For fields where you enter information, this ***must*** be blank.

  For information only fields, such as screen titles, this field holds part or all of the screen text, and you can change it as desired.

- Field Default: This is the default entry for the field. You can change it, but be sure your entry is a legal entry for this field in AlphaCALC.

- Pop-Up Menu Selections: Only fields with a limited number of responses (for example, cell alignment) have entries here. When using the field, you can make an entry by typing the first letter of one of these selections. You may change the wording of the choices only (for example to translate into another language). Each selection should start with a different letter.

Do NOT add entries, delete entries, or change the order of the entries in this field. AlphaCALC expects to see exactly these options in exactly this order. If you change the order, your selections in AlphaCALC will not have the effect you expect.

- Field Prompt: This is the screen prompt for the field. You may change it as desired.

- On-Line Help: This is the message which appears at the bottom of the screen when the cursor is on this field in AlphaCALC. You can change the message portion as desired.

  At the end of this field is the | character, followed by a file name. This is the name of the help file displayed when [HELP] is pressed at this field. Do not remove the | character or change this file name!

4. Change all the fields you want, obeying these instructions and warnings. After making your changes, press [EXECUTE]. You return to the Screen Editor display.

If you changed a field by mistake, press [MENU] instead of [EXECUTE] to return to the Screen Editor.

5. After you've changed all the fields you want, press [MENU] from the Screen Editor to return to the Main Menu.

6. To save your changes, use either the **SAVE** or **FINISH** command.

Changing the Screen Title

Many AlphaCALC screens have a screen title in reverse video at the top of the screen. Because these titles are "laid over" a box already drawn on the screen, the modification process is a little more complicated.

If you follow the process above for the screen title, instead of seeing the Field Editor screen for the title, you see the information defining the box. So, to modify the title, you need to move the box off the title, change the title, then move the box back. To do so:

1. Use ALCSCR to call up the screen, as described above.

2. Move the cursor to the top left corner of the box, and press [BLOCK MARK] or [CTRL] / [P].

3. Move the cursor down one line and press [ESC] [M].

4. This moves the top of the box down one line. Now, follow steps 1 - 4 above to change the title text.

5. Move the cursor to the top left corner of the box. Press [BLOCK MARK] or [CTRL] / [P].

6. Move the cursor up one line (back to the original box position) and press [ESC] [M].

7. Press [MENU] to return to the Main Menu, and use **SAVE** or **FINISH** to save your changes.

SCREEN FILE NAMES

The following lists give the name and description of each screen file used by AlphaCALC. The first list gives the files in the ALC: account; the second gives the files in the DMSESP: account. The original, American English screens have the extension .USA.

ALC: File	Description
ACSACM	ACUM function input screen
ACSADV	AUTO ADVANCE directions screen
ACSCEL	Cell editing screen (bottom of worksheet display)
ACSCMD	Editing Menu
ACSCPY	**COPY** command screen
ACSCRE	**CREATE** command screen
ACSCUT	**CUT** command screen
ACSCVT	Converting worksheet message
ACSDIR	**DIRECTORY** options screen
ACSDRH	History directory display
ACSDRN	Name directory display
ACSDRS	Standard directory display
ACSDUP	**DUPLICATE** command screen
ACSED2	**EXPORT** command file loading screen
ACSEDT	**EDIT** command screen
ACSEPO	**EXPORT** command screen
ACSERA	**ERASE** command screen
ACSEWV	Erase worksheet verification prompt
ACSGRF	GRAPH SETUP screen
ACSGTO	GOTO CELL screen
ACSHDR	Worksheet header
ACSHL1	Main Menu Help Menu
ACSHL2	Editing Menu Help Menu
ACSHL3	Worksheet display Help Menu
ACSIPO	**IMPORT** command screen
ACSITR	RECOMPUTE iteration options screen
ACSLAM	Loan amortization schedule
ACSLAN	Loan amortization analysis
ACSLAO	Loan analysis options
ACSLOD	Loading worksheet message
ACSLON	Loan schedule/analysis selection
ACSLSO	Loan schedule options
ACSMNU	Main Menu
ACSMOD	**MODE** command screen
ACSOVW	Overwrite file verification screen
ACSPAG	**PAGINATE** command screen
ACSPBR	Page break setting screen
ACSPRN	Main Menu **PRINT** command screen
ACSPRO	**PROTECT** command screen

ALC: File	Description
ACSPRT	Editing Menu **PRINT** command screen
ACSPST	**PASTE** command screen
ACSQUE	Printer queue maintenance screen
ACSQWT	Editing Menu **QUIT** verification prompt.
ACSREN	**RENAME** command screen
ACSREP	**REPLACE** and **GLOBAL** command screen
ACSSCH	**SEARCH** command screen
ACSSET	**FORMAT** command screen
ACSSRT	**SORT** command screen
ACSWFL	Slash file command menu
ACSWPR	Slash print command menu
ACSWRG	Slash range command menu
ACSWRT	**WRITE** command screen
ACSWSC	Slash command menu
ACSWWS	Slash worksheet command menu

These are the files in the DMSESP: account:

DMSESP: File	Description
DASMAC	Macro screen
GRFSEL	Graphics device selection
OPNBOX	Directory from command screens
PRTSEL	Printer selection
PSSWRD	Password entry
SEHEDT*	Help for ALCSCR editor
SESCMD	Screen editor command screen
SESFLN	Field editor screen (80 column)
SESFLW	Field editor screen (132 column)

* SEHEDT is a help screen. Follow the help screen procedure for changing this file.

CHANGING AlphaCALC'S HELP FILES

AlphaCALC has approximately 60 help files, one for each help topic available on the three AlphaCALC Information Menus. Each help file is a formatted text file containing display codes which AlphaCALC interprets as terminal display attributes.

You can either modify these help files, or use them as a guide for creating your own, translated help files.

PROCEDURE FOR CHANGING THE HELP FILES

AlphaCALC's help files are standard AMOS text files you can modify using AlphaVUE. Special control codes, corresponding to TCRT calls, are embedded in the text to control the special screen attributes we use, such as underlining.

Whether you copy the provided help files, or create your own, here are some procedures you should use:

- If you modify the existing files, ***make sure*** you save a copy of the original files. Don't risk losing the original help information if you accidentally erase part or all of the file you're working on.

- You can use AlphaVUE's Control-G feature, as described in the *AlphaVUE User's Manual*, to insert the screen display control codes in the text. The codes consist of an [ESC] (which appears as ^[) followed by a number. The codes we use in the help files are:

11	Start dim display
12	Start bright display
30	Start underline
31	End underline
32	Start reverse video
33	End reverse video

- If you create your own files "from scratch," you may want to use TXTFMT to format the files. If you do, you should set the page dimensions to match the size of the screen, so one help screen equals one page in the formatted file.

- While you do not have to use the same screen format and conventions we do, we strongly suggest you do adhere to a consistent set of conventions throughout your help files.

HELP FILE NAMES

AlphaCALC's help files are located in ALC:. Their names indicate the Information Menu topic they belong to. The original, American English, help files have the extension .USA.

The help files have names in the format CH?XXX.USA, where:

- CH indicates this is an AlphaCALC help file.
- ? indicates which Information Menu the file is for. M stands for Main Menu, E for Editing Menu, and W for the worksheet display.
- XXX is a three digit number indicating the topic number.

For example, the file CHE006 is for help topic number 6 on the Information Menu you can call up from the Editing Menu.

The lists below give the name and the topic covered by each help file:

Main Menu

Topic	File Name	Description
1	CHM001	Getting Help
2	CHM002	Using Menus and Command Screens
3	CHM003	CREATE - Creating a New Worksheet
4	CHM004	EDIT - Editing an Existing Worksheet
5	CHM005	DIRECTORY - Listing Worksheets
6	CHM006	PRINT - Printing a Worksheet
7	CHM007	QUEUE - Viewing the Print Queue
8	CHM008	COPY - Copying a Worksheet
9	CHM009	ERASE - Erasing a Worksheet
10	CHM010	RENAME - Changing a Worksheet Name
11	CHM011	MODE - Changing Operation Options
12	CHM012	IMPORT - Importing Worksheet Data
13	CHM013	EXPORT - Exporting Worksheet Data
14	CHM014	LOAN - Loan Amortization
15	CHM015	QUIT - Leaving AlphaCALC

Editing Menu

Topic	File Name	Description
1	CHE001	Using Menus and Command Screens
2	CHE002	FINISH - Leave Your Worksheet
3	CHE003	SAVE - Saving Your Worksheet
4	CHE004	QUIT - Abandon Your Changes
5	CHE005	SEARCH - Looking for an Entry
6	CHE006	REPLACE/GLOBAL - Replacing an Entry
7	CHE007	HEADER - Worksheet Header Information
8	CHE008	DUPLICATE - Copying a Cell Range
9	CHE009	PROTECT - Protecting a Cell Range
10	CHE010	FORMAT - Setting Cell Format
11	CHE011	MODE - Changing Operation Options
12	CHE012	PAGINATE - Paginating Your Worksheet
13	CHE013	PRINT - Printing Your Worksheet
14	CHE014	QUEUE - Viewing the Printer Queue
15	CHE015	WRITE - Creating an AlphaWRITE Document
16	CHE016	DIRECTORY - Listing Worksheets
17	CHE017	SORT - Sorting Data
18	CHE018	WIDTH - Changing Display Width
19	CHE019	CUT - Moving Data to Another Worksheet
20	CHE020	PASTE - Copying Data from Another Worksheet
21	CHE021	IMPORT - Importing Worksheet Data
22	CHE022	EXPORT - Exporting Worksheet Data
23	CHE023	LOAN - Loan Amortization

Worksheet Display

Topic	File Name	Description
1	CHW001	Moving Around the Worksheet
2	CHW002	Entering and Editing Cell Formulas
3	CHW003	Types of Cell Formulas
4	CHW004	AlphaCALC Functions
5	CHW005	Displaying Row and Column Borders
6	CHW006	Locking Rows and Columns
7	CHW007	Moving Rows and Columns
8	CHW008	Marking and Clearing Blocks of Cells
9	CHW009	Copying a Block of Cells
10	CHW010	Erasing a Block of Cells
11	CHW011	Inserting and Deleting Rows and Columns
12	CHW012	Erasing Rows and Columns
13	CHW013	Recomputing Your Worksheet
14	CHW014	Graphics
15	CHW015	Protecting Cells
16	CHW016	Automatic Advance
17	CHW017	Inserting Page Breaks
18	CHW018	Macro Record and Playback
19	CHW019	Universal Key Sequences
20	CHW020	Using Slash Commands

MODIFYING THE MESSAGE FILE

AlphaCALC's error and status messages are in the file ALCMSG.USA. This is an assembled version of the source file ALCMSG.M68. To translate these messages, or change the messages for American English, you need to modify the ALCMSG.M68 file, then reassemble it.

ALCMSG.M68 is an assembly language program source file. Before starting this procedure, be sure to have a backup copy of the file. Be very careful while modifying it, and do not change any part of the file except those described here. If you do, you could cause serious problems or disable AlphaCALC altogether.

CHANGING THE MESSAGES

To modify ALCMSG.M68, log into ALC: and call it up using AlphaVUE or AlphaXED.

Comments in the ALCMSG.M68 file indicate which section of the file contains the messages you can change. The messages are near the middle of the file, on lines containing the macro MSGTXT. The actual text is contained in angle brackets following MSGTXT. For example:

```
CE.SNF:  MSGTXT  <Specified string was not found>
```

You can modify the message text any way you want, making it longer or shorter if necessary, but be sure it is enclosed with angle brackets, and ***do not change any other part of the line***.

Some messages have a second set of angle brackets, containing codes which begin DE$. Do not change these codes! They control when and how the messages display.

TRANSLATING THE MESSAGE FILE

If you want to create a version of the message files for a language other than American English, you must make one other change to ALCMSG.M68.

Near the beginning of the file is a line like this:

```
OBJNAM 0, 0, [USA]
```

This determines the extension of the assembled message file. Change the USA to the extension for the language you want.

ASSEMBLING THE MESSAGE FILE

Once you've made all your changes to ALCMSG.M68, you must assemble it. You can do this in either of two ways:

- Use the AlphaVUE or AlphaXED GO command.
- From AMOS command level, enter the command:

  M68 ALCMSG RETURN

Either of these methods assembles the message file, giving the resulting object file the extension specified by OBJNAM.

Some messages you see in AlphaCALC are sent by the ESP toolbox. Depending on the version of ESP, you may or may not be able to change these messages. Refer to the ESP documentation.

CHAPTER FIVE
GRAPHICS SETUP

AlphaCALC enables you to create five types of graphs: line graphs, stacked or clustered bar charts, area charts, and pie charts. You can either display these on a graphics terminal or print them to a graphics capable printer.

To set up the graphics option, you need to do the following:

- Make sure Alpha Micro's Graphics Operating System (AMIGOS) software is installed correctly on your computer.
- Modify your system initialization file to load the files GRAPH.SYS and AMIGOS.SYS into system memory. You also need to load the graphics driver programs for the graphics devices you want to use.
- Define the available graphics devices to AlphaCALC in the AlphaCALC initialization file.
- You may also need to adjust the memory area AlphaCALC reserves for use with graphics (this affects the maximum number of points you can include on a graph). You can do this with an AlphaCALC initialization file command.

This chapter discusses each of these areas.

INSTALL AMIGOS

AMIGOS is Alpha Micro's graphics operating system. AlphaCALC uses the AMIGOS software to create its graphics. If AMIGOS is not already installed on your computer, you can get it from your Alpha Micro dealer. Follow the AMIGOS installation instructions before continuing with AlphaCALC's graphics setup.

Some of the steps below repeat information and instructions contained in the AMIGOS installation process. This is to make sure you don't omit any steps crucial to AlphaCALC's graphics procedures.

MODIFYING THE SYSTEM INITIALIZATION FILE

As part of the AMIGOS or AlphaCALC installation process, you should have added statements to your initialization file loading GRAPH.SYS and AMIGOS.SYS into system memory. Somewhere in the SYSTEM section of the initialization file, you should see these lines:

```
SYSTEM GRAPH.SYS
SYSTEM AMIGOS.SYS/N
```

If these lines are not in your system initialization file, add them.

As always, do **not** modify your system initialization command file directly. Always copy the file and modify the copy! MONTST using the test copy after making your changes, and RENAME or COPY the test file only after you know the changes are correct.

For AlphaCALC to generate a graph, the graphics driver for the device you're generating the graph on must be in either system or user memory. If you want, you can require each AlphaCALC user to load the appropriate driver into user memory before calling up AlphaCALC. However, it is generally easier, and more efficient if more than one person will be creating graphs, to load the drivers into system memory.

The graphics drivers are in the DVR: account (DSK0:[1,6]), and have the extension .GDV. For a graphics printer, the file name is the same as the printer driver name; for terminals, it is the same as the terminal driver name. For example, the graphics driver for an AM-72 terminal is AM72.GDV.

Determine what devices (terminals and printers) you may want to generate AlphaCALC graphs on, and add a SYSTEM statement for each one to your initialization file. For example:

```
SYSTEM DVR:AM72.GDV
SYSTEM DVR:AM62G.GDV
SYSTEM DVR:LASWRT.GDV
SYSTEM DVR:LASLND.GDV
SYSTEM DVR:LASPOR.GDV
```

There are two special graphics drivers for PostScript laser printers, LASLND.GDV and LASPOR.GDV, in addition to LASWRT.GDV. The LASLND driver allows you to print graphs in landscape mode. LASPOR prints in portrait mode, but generates slightly larger graphs than LASWRT.GDV.

DEFINING THE AVAILABLE GRAPHICS DEVICES

Just as you define printers to AlphaCALC in either the AlphaCALC initialization file or the system PRTDEF file, you need to define the available graphics devices in either the AlphaCALC initialization file or the system graphics definition file, SYS:GRFDEF.lan (where lan is the file extension for your language).

AlphaCALC follows the same rules when looking for graphics definitions as it does looking for printer definitions. If there are any graphics devices defined in your AlphaCALC initialization file, those definitions are used. The GRFDEF file definitions are used only when no graphics devices are included in the AlphaCALC initialization file.

Any printer you want to define as a graphics device **must** also be defined to AlphaCALC as an available printer.

The statement to use to define a graphics device is GRAPHICS-DEVICE. This is the format:

```
GRAPHICS-DEVICE=Graphics Driver,Printer Name,"Name"
```

Here are two sample commands, one defining a terminal, the other a printer:

```
GRAPHICS-DEVICE=AM72,,"AM-72 color graphics terminal"
GRAPHICS-DEVICE=LASWRT,LASWR1,"The Laser Printer"
```

This is what each part of the line means:

- Graphics Driver: The name of the graphics driver program for this device. Do not include the .GDV extension.

- Printer Name: For graphics printers, you must include the name by which the printer is defined to AlphaCALC, in either the AlphaCALC initialization file or PRTDEF. The printer sample above is for the printer named LASWR1.

  For terminals, leave this argument out, as shown in the first example, but be sure to include two commas after the driver name.

- Name: This is the description which will display on your screen when you choose the device you want. It must be enclosed in quotes, as shown, and can be up to 30 characters long.

You may want to put the most commonly used device first in the initialization file, since it will be the default on the device selection screen.

For more information about the AlphaCALC initialization file, including how to create different files for different languages and different accounts, see Chapter 3.

THE MAX-GRAPH-POINTS COMMAND AND MEMORY USAGE

When creating a graph, AlphaCALC must be able to hold the data for all the points on the graph in user memory. Therefore, when you press the [GRAPH SETUP] or [GRAPH DISPLAY] key, AlphaCALC reserves a section of user memory to hold the graph data. The size of this area is set by the MAX-GRAPH-POINTS command.

MAX-GRAPH-POINTS sets the maximum number of data points you can graph in a single data range. Since you can enter up to eight data ranges, the maximum total number of points on a graph is eight times the MAX-GRAPH-POINTS setting.

When you press GRAPH SETUP or GRAPH DISPLAY, AlphaCALC reads the MAX-GRAPH-POINTS setting and reserves a portion of user memory large enough to hold the maximum amount of data for the graph. This area of memory is then unavailable for any other use (such as to hold worksheet data).

The default setting for MAX-GRAPH-POINTS is 40 data points per range. At this setting, AlphaCALC reserves approximately 20K of memory for graph data. Each point you raise this setting adds 464 bytes to the area needed. For example, if MAX-GRAPH-POINTS = 50, the memory area reserved is between 24K and 25K. Of course, lowering the maximum number of points reduces the memory requirement the same amount.

Area and stacked bar charts cannot have more than 40 points per data range regardless of the MAX-GRAPH-POINTS setting.

CHAPTER SIX
AlphaCALC'S MEMORY USAGE

The *Release Notes* for your version of AlphaCALC list the memory partition size required for each person on your computer who wants to use AlphaCALC. This chapter discusses exactly what that requirement means, how you can reduce the memory needed for each user, and why you may want to increase the partition size depending on how you'll be using AlphaCALC.

WHAT AlphaCALC USES MEMORY FOR

When you're using AlphaCALC, your memory partition contains the following memory modules:

Module	Size	Description
....	10K	Reserved for AMOS use; holds disk drivers, etc.
CALC.LIT	32K	AlphaCALC's controlling program
CALC.INI	5K	CALC.INI file settings, cell formats
CALC.IMP	120K	Impure area—used for program overlays, work area for calculations, work area for graphics, etc.
SCREEN.MEM	15K	Screen files, help files.
CALC.MEM	Rest of partition (at least 32K)	Worksheet data, work area for SORT, printer drivers.

These sizes are all approximate.

REDUCING AlphaCALC's MEMORY USAGE

There is only one way to substantially reduce the amount of user memory AlphaCALC needs—load CALC.LIT in system memory. This saves about 32K of memory space per user, while adding 32K to the system memory size.

Unlike many other Alpha Micro software packages, there is no benefit to loading AlphaCALC's overlays in system memory. A space (approximately 10K) is reserved at all times in the user impure area to load overlays. If the overlay is in system memory, it isn't loaded, but the reserved space is still unavailable for any other use.

You can also achieve minor memory savings by loading frequently used printer drivers (which are loaded in the AlphaCALC data area in memory, not the impure area) in system memory.

Graphics device drivers must be loaded either in system memory or in the user's memory partition ***before*** AlphaCALC is called up. See Chapter 5.

ESP TOOLBOX MEMORY USE

As described in Chapter 2, you must load the ESP toolbox file, TOOLBX.SYS, in system memory when you install AlphaCALC. This file occupies about 85K of disk space, but may take up much more room in user memory. There are two reasons for this: the per user memory area, and the memory pool.

The following sections assume you do not have ESP installed on your system, and use the toolbox only for AlphaCALC. If you do use ESP, keep the ESP requirements in mind when setting the toolbox parameters.

Unless you're an experienced ESP user, do not make any changes to TOOLBX.INI other than those described below. You could adversely affect AlphaCALC's operation if you do.

TOOLBOX PER USER MEMORY

In addition to the 85K of program code space, when TOOLBX.SYS is loaded in system memory, it reserves a space of approximately 10.5K for each user, which it uses to hold printer and graphics device definitions (up to 32 of each per job), job characteristics, and other information. Obviously, on a system with lots of users, this can take up a lot of memory.

When figuring how many users to reserve space for, the toolbox looks in two places:

1. If there is a USERS=X statement in the TOOLBX.INI file in DMSESP:, it reserves space for that many users.

2. If there is no USERS statement, the toolbox counts the number of active jobs on the system—ignoring task manager or printer spooler jobs, etc..

If not everyone on your computer will be using AlphaCALC, you can reduce the size of the toolbox in system memory by setting USERS to the correct number. To take an extreme example, if your computer supports 200 users, but only ten will be using AlphaCALC, you can save about 2 megabytes of memory space by setting USERS=10 in TOOLBX.INI.

If you do add or change the USERS command, you must reboot your system for the change to take effect.

SORT AND THE TOOLBOX MEMORY POOL

The toolbox memory pool is an area set aside in TOOLBX.SYS. Like the number of users, the size of the memory pool is set in DMSESP:TOOLBX.INI. The only part of AlphaCALC which may use the memory pool is the **SORT** command.

As mentioned above, **SORT** uses space in AlphaCALC's data area as temporary storage. However, if **SORT** runs out of room in the user's partition, it uses the toolbox memory pool.

If you frequently perform large sorts, and they seem to take longer than they should, it may be because **SORT** is running out of room in both user memory and the memory pool. When this happens, **SORT** is forced to write data to the disk and read it back during the sort, slowing the process considerably.

If you think this might be the case, you can either increase the partition size for everyone who uses **SORT**, or increase the size of the toolbox memory pool.

To change the memory pool size, log to DMSESP: and use AlphaVUE to modify TOOLBX.INI. Find the line which reads `MEMORY=xxK` and change the number after the equal sign to the size you want.

After changing the MEMORY command, you must reboot your system for the change to take effect.

On the other hand, you may be able to save some room in system memory by reducing the memory pool size if you are not having any problems with **SORT**.

CHAPTER SEVEN
FILES AlphaCALC CREATES

As the AlphaCALC system administrator, you probably already know that AlphaCALC worksheets are stored in disk files with the name you gave the worksheet and the extension .ALC. But, you may not be familiar with all the other kinds of files various AlphaCALC features can create. This chapter lists all these type of files, giving the extension used for each, when they're created, and a brief description of the contents.

All of the files AlphaCALC creates are located in the disk account you called AlphaCALC up from.

BACKUP FILES

Every time you **SAVE** or **FINISH** a worksheet, the current version of the worksheet is saved with the extension .ALC. But, the previous version of the worksheet is also saved, and given the extension .BAK. So, at any given times, there are two versions of a worksheet saved on disk, the .ALC file and the .BAK file.

If anything happens to the .ALC file—for example, if you accidentally erase it or save changes you don't really want to keep—you can go to AMOS command level and rename the .BAK file with a .ALC extension. This returns the worksheet to a "one version back" state, and you can re-enter any changes you need.

When you rename a .BAK file, do ***not*** erase the .ALC file you're replacing until you're sure the .BAK file is the version of the worksheet you want. When you use the AMOS RENAME command, give the .ALC file a temporary name, such as OLD.ALC, until you can check its contents.

The .BAK file is not deleted when you use the **ERASE** command to erase a worksheet. Therefore, if you're erasing a worksheet for security reasons, you may want to go to AMOS command level and erase the .BAK file. Otherwise, someone may be able to rename the .BAK file to access the worksheet data.

PRINT FILES

When you use **PRINT** from either the Main or Editing Menu, AlphaCALC creates a disk file to hold the information you want printed. Depending on the choices you make on the printer selection screen, this file may be sent to the printer immediately and erased after printing, printed and kept on the disk, or just created on the disk and not sent to a printer.

The print file is given the same name as your worksheet, and the first available extension in the numeric series .000, .001, .002 and so on. For example, if you print a worksheet named MNTHLY, and the file MNTHLY.000 already exists, the file **PRINT** creates has the name MNTHLY.001.

The exact contents of the print file depend on the printer driver you use to create it. For example, if you print to a printer which uses the MISC printer driver, the print file looks almost the same as your worksheet. On the other hand, if you print to a PostScript™ laser printer using a printer driver such as LASWRT.PDV, the many formatting commands included in the file make it almost unreadable unless you know PostScript format.

DIRECTORY PRINT FILES

The **DIRECTORY** command follows basically these same rules when you print. The file can be sent to the printer and erased, printed and kept, or just stored on the disk, as with **PRINT**. Again, the format of the file depends on the printer driver used by the printer you selected.

The file name of a **DIRECTORY** print file is always DIRECT. The extension is the first available in the series .000, .001, .002 and so on, as described above.

EXPORT FILES

The **EXPORT** command can create files containing data in four different formats. You have complete control over the file name and extension for any file you create using **EXPORT**. However, these are the default extensions used if you do not specify one:

FORMAT	EXTENSION
DIF	.DIF
SYLK	.SLK
Cell ID, formula	.CID
Row	.ROW

For a description of the formats of these files, see the *AlphaCALC Programmer's Information Manual*.

GRAPH FILES

When you create a graph using the [GRAPH SETUP] key, the graph parameters are stored in a file with the same name as the worksheet and the extension .GDF. When you print a graph, a print file is created as described above in the "Print Files" section.

PostScript is a trademark of Adobe Systems Incorporated.

When you edit an existing graph, AlphaCALC creates a backup file for the .GDF file (as it creates a .BAK file when you edit your worksheet). This file has the extension .GDB.

AlphaWRITE DOCUMENT FILES

AlphaCALC's **WRITE** command creates an AlphaWRITE document from all or part of your worksheet. You can give this document any name you choose, and as with all AlphaWRITE documents, its extension is .WRT.

The **WRITE** command creates an AlphaWRITE 1.2A format document. This is done to ensure the greatest possible compatibility. If you have AlphaWRITE 2.0 or later, the document will be converted to the proper format automatically when you call it up.

MACRO FILES

A macro you create using the [MACRO] key is stored in a file with the name you gave the macro and the extension .TIP.

CHAPTER EIGHT
EXTENDED CHARACTER SUPPORT

AlphaCALC supports the full ISO Latin I character set. This is also called "eight-bit" character support, since each character is comprised of eight bits instead of the seven bits required for an ASCII character. ASCII characters have values from 0 to 127; numeric values for ISO Latin I characters go up to 255. The characters with values from 128 to 255 are sometimes called "extended" characters.

You can use any extended character in AlphaCALC. However, most keyboards do not have keys for these characters. So, how can you enter them?

AlphaCALC does not include any special software to allow you to enter extended characters. To use extended characters with AlphaCALC, you must use a terminal and terminal driver which support hardware composition of extended characters.

Two Alpha Micro terminals that support hardware composition are the AM-65 and the AM-75. The terminal drivers for these terminals to use extended characters are AM65EX.PDV and AM75EX.PDV. You must use these terminal drivers in order to use extended characters in AlphaCALC with these terminals.

For how to enter extended characters on these terminals, see your terminal owner’s manual. The procedure varies depending on the terminal type, the kind of keyboard you have and so on.

APPENDIX A
FILE LIST

The following table lists all the types of files which make up the AlphaCALC software. During installation, all these files are copied to the ALC: account. Several of them are then copied to other locations, and/or renamed, as described in Chapter 2.

FILE NAME	DESCRIPTION
*.DIR	CALC, ESP and DART directories
CALC.LIT	Main AlphaCALC program
CALC.NEW	Prototype AlphaCALC initialization file
ALC*.OVR	Program overlays
ALCSCR.LIT	Screen editing program
*.CAX	Function key translation tables
AMSORT.*	AMOS sorting program
BASORT.*	BASIC interface for AMSORT
WRITE.PDV	Printer driver for WRITE command
EXTCEL.*	AlphaBASIC and assembly language interfaces
ACS*.USA	AlphaCALC screen files
CH*.USA	Help files
DAC001.USA	Desk accessory menu (not used)
DASMAC.USA	Macro screen
PRTSEL.USA	Printer selection screen
GRFSEL.USA	Graphics device selection screen
SE*.USA	Screen editor screen display files
ALCMSG.M68	Source file for messages
MSGSYM.M68	Symbol file for messages
PSSWRD.USA	Password entry screen
TBXMSG.USA	Toolbox message screen
TOOLBX.SYS	ESP toolbox
TOOLBX.NEW	Prototype toolbox initialization file

DOCUMENT HISTORY

REVISION	DATE	DESCRIPTION
00	10/3/91	New document for AlphaCALC 2.0 release.

INDEX

.000 files 7-1
.BAK files 7-1
.CID files 7-2
.DIF files 7-2
.GDB files 7-2
.GDF files 7-2
.ROW files 7-2
.SLK files 7-2
.TIP files 7-3
.WRT files 7-3

A

Absolute cell reference
  changing designator 3-3
ABSOLUTE-INDICATOR command 3-3
ACALC/P command 2-10
ALC: account 2-2, 2-6
ALCMSG.M68 file 4-9
ALCSCR program 4-2 to 4-5
  Field Editor screen 4-3
AlphaCALC initialization file
  BORDERCOLOR command 3-3
  character setting commands 3-2
  command format 3-2
  CUBE-WARNING command 3-3
  default 3-2
  FORMAT-MASK command 3-5 to 3-7
  GRAPHICS-DEVICE command 3-4, 5-3
  language specific 3-1
  MAX-GRAPH-POINTS command 3-4, 5-3
  names and locations 3-1
  precedence 3-1
  PRINTER command 3-4
AlphaWRITE document files 7-3
AMIGOS 5-1
  installation 5-1, 5-2
AMOS 1.X version
  copy commands 2-4, 2-8
AMOS 2.X version
  copy commands 2-4, 2-8
AMOSL.INI, AMOS32.INI — see System Initialization File
AMSORT.SYS 2-4, 2-8
Audience
  intended 1-1

B

Backup graphics files 7-2
Backup worksheet files 7-1
BASORT.SBR 2-4, 2-8
BORDERCOLOR command 3-3

C

CALC.DIR file 2-2, 2-7
CALC.INI file — see AlphaCALC initialization file
CALC.lan file 3-1
CALC.LIT
  loading in system memory 6-1
  renaming for dual version installation 2-9
CALC/P command 2-5
Cell formats
  defining 3-5 to 3-7
Cell range
  changing separator 3-3
Character setting commands
  AlphaCALC initialization file 3-2
Colors
  setting for worksheet border 3-3
Command format
  AlphaCALC initialization file 3-2
Compatibility with DART/ESP 2-3, 2-8
Conventions used in this book 1-2
CPUID in PRINTER command 3-5
CUBE-WARNING command 3-3

D

DART.DIR file 2-3, 2-8
DART/ESP
  checking compatibility 2-3, 2-8
Defining printers 3-4
  over AlphaNET 3-5
Desk accessories in TOOLBX.INI file 2-11
DIRECT.000 files 7-2
Directory file (CALC.DIR) 2-2
Directory print files 7-2
Display attributes used in help files 4-7
DMSESP: account 2-2, 2-6
Downloading the AlphaCALC software 2-2, 2-7
Drivers
  graphics device 5-2
Dual version installation 2-6 to 2-10

E

Erase command and backup files 7-1
Ersatz names 2-2, 2-6
ESP
  checking compatibility 2-3, 2-8
ESP toolbox
  memory pool and sort command 6-3
  system memory requirements 6-2
ESP.DIR file 2-3, 2-8
Export files
  default extensions 7-2
Extended character support 8-1

F

File extensions
  export defaults 7-2
Files
  created as worksheet backup 7-1
  created by write command 7-3
  created during directory printing 7-2
  created during export 7-2
  created during graphics 7-2
  created during macros 7-3
  created during printing 7-1
  included on release tape A-1
Foreign language support
  AlphaCALC initialization file 3-1
  screen, message, and help file translation 4-1
FORMAT-MASK command 3-5 to 3-7

G

Graphic conventions 1-2
Graphics
  defining devices to AlphaCALC 5-2
  determining memory requirements 5-3
  files 7-2
  loading drivers in memory 5-2
  setup overview 5-1
GRAPHICS-DEVICE command 3-4, 5-3
GRFDEF file 5-2

H

Help files
  display attributes used in 4-7
  list 4-8
  modifying 4-6 to 4-8
  name format 4-7

I

Installation
  dual version 2-6 to 2-10
  overview 2-1
  single version 2-1 to 2-5
Installing graphics
  overview 5-1
ISO Latin I support 8-1

L

LAYER-SEPARATOR command 3-3

M

Macro files 7-3
Macros
  setup in TOOLBX.INI file 2-11
MAX-GRAPH-POINTS command 3-4, 5-3
MEMORY command
  in TOOLBX.INI file 6-3
Memory requirements 6-1, 6-2
  changing for graphics 5-3
  for ESP toolbox 6-2
  reducing 6-1
  sort command and ESP memory pool 6-3
Modifying AlphaCALC's help files 4-6 to 4-8
Modifying AlphaCALC's messages 4-9
Modifying AlphaCALC's screen displays 4-2 to 4-5
MONTST command 2-5, 2-10
Multiple layer operations
  setting warning message 3-3

O

Other reference material 1-2
OVERFLOW character command 3-3
Overlay files memory use 6-2

P

Print files 7-1
PRINTER command 3-4
  CPUID 3-5
  format 3-4
  printer description 3-5
  printer driver name 3-5
  printer name 3-5
Printers
  defining 2-5, 2-10
  defining as graphics device 5-3
Product Installation Code (PIC) 2-5, 2-10
PRTDEF file 3-4

R

RANGE-SEPARATOR command 3-3
Reference material
  other manuals 1-2
Remote printers
  defining 3-5
Round digits
  in format definitions 3-6

S

Scaling factor
  in format definitions 3-7
Screen files
  list 4-5
  name format 4-2
Screen titles changing 4-4
Screens, modifying 4-2 to 4-5
Single version installation 2-1 to 2-5
Software files
  copying to correct accounts 2-3, 2-8
  list A-1
Sort command memory use 6-3
System initialization file 2-4, 2-9
  AMIGOS setup 5-2
  removing SYSTEM statements 2-10
  SYSTEM statements for graphics devices 5-2
  SYSTEM statements required 2-4, 2-9
System memory
  ESP toolbox size 6-2
  loading CALC.LIT 6-1
System printer definition file (PRTDEF) 3-4

T

Terminal drivers
  needed for extended characters 8-1
Terminals
  defining as graphics device 5-3
Toolbox
  system memory requirements 6-2
TOOLBX.INI file
  changing ESP memory requirements 6-2
  MEMORY command 6-3
  setup for ESP users 2-11
  USERS statement 6-2
TOOLBX.SYS file
  system memory size 6-2

U

User memory
  contents and requirements 6-1, 6-2
USERS statement
  in TOOLBX.INI file 6-2

V

VCRRES command 2-2, 2-7
VERIFY command 2-2, 2-3, 2-7, 2-8

W

Worksheet borders
  setting color 3-3